

Line sensor

DETECT

INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



Warehouse guide-line robot

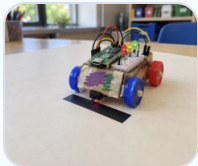
DETECT
Read guide-line position



DECIDE
Is the vehicle drifting?



DO
Correct the steering



Cardboard lap-counting car

DETECT
Cross dark finish strip



DECIDE
Is this a new lap?



DO
Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Code

● Raspberry Pi

```
from gpiozero import LineSensor  
  
line = LineSensor(16)  
  
on_line = line.is_active
```

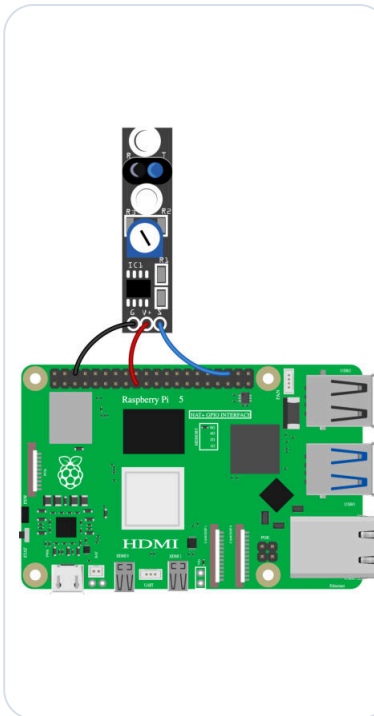
● Raspberry Pi Pico

```
from picozero import DigitalInputDevice  
  
line = DigitalInputDevice(22, pull_up=False)  
  
on_line = line.is_active
```

Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

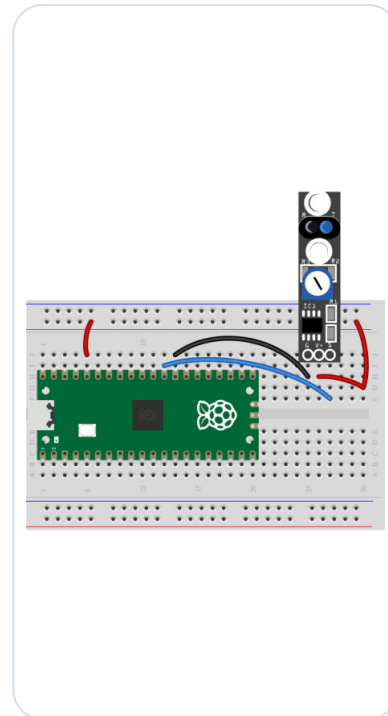
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → V+, **GPIO 16** → S

PICO



GND → GND, **3V3 OUT** → V+, **GP22** → S